

Longitudinal Multi-view Data Analysis

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Assignment 1

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Overview

This report addresses three statistical questions involving a contingency table that comes from the built-in R dataset ``asbestos``.

This dataset cross-classifies patients' asbestosis grades (columns) with years of exposure (rows). The primary focus includes:

- Testing independence using different test statistics (Pearson's Chi-squared, Monte Carlo variants, Freeman–Tukey, and Log-Likelihood Ratio).
- Exploring the family of Cressie–Read divergence measures.
- Evaluating an asymmetric association measure (Goodman–Kruskal's tau).

Data Table

We use the following 4×4 data from the ``asbestos`` dataset. The row variable is years of exposure (Y), and the column variable is asbestosis grade (G). The table is as follows:

	G1	G2	G3	G4
Y1	5	8	6	7
Y2	9	13	12	7
Y3	11	20	7	6
Y4	4	6	6	4

In R, this table is defined as ``asbestos``.

Question 1: Chi-squared Test and Cressie–Read Statistics

- Use the standard χ^2 test (chisq.test) to test for independence.
- Compare Monte Carlo p-values with different simulation sizes (B = 10, 100, 1000, 10000).
- Manually compute chi-squared statistic and Cressie–Read divergences for $\lambda = 0, 0.5$, and 1.

Hypotheses

H0: Y and G are independent.

H1: Y and G are not independent.

Degrees of freedom = $(4-1)(4-1) = 9$

1. Standard Pearson's Chi-squared Test

Running the test in R:

```
res_chi <- chisq.test(asbestos)
```

```
res_chi
```

We obtain:

- $\chi^2 \approx 33.882$
- df = 9
- p-value ≈ 0.0002073

This small p-value (< 0.05) leads us to reject H_0 . There is significant association between rows and columns.

:Pearson's Chi-squared

$$\sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \chi^2$$

:Cressie–Read

$$O_{ij} \left[\left(\frac{O_{ij}}{E_{ij}} \right)^\lambda - 1 \right] \sum_{i,j} \frac{2}{\lambda(\lambda+1)} = I_\lambda(O, E)$$

2. Monte Carlo p-values

Using `simulate.p.value = TRUE` with different numbers of simulations B:

- B = 10 → p-value ≈ 0.1818
- B = 100 → p-value ≈ 0.01
- B = 1000 → p-value ≈ 0.001
- B = 10000 → p-value ≈ 0.0001

As B increases, Monte Carlo estimates stabilize and confirm the significance of association.

Manual Chi-squared Computation

Let E_{ij} be the expected frequency under independence. Then:

In R:

```
expected <- chisq.test(asbestos)$expected
observed <- asbestos
chi_manual <- sum((observed - expected)^2 / expected)
chi_manual
```

We get:

$$\chi^2_{\text{manual}} \approx 33.879$$

Which matches the built-in test result.

3. Cressie–Read Divergence

We evaluate $\lambda = 0$ (LLR), 0.5, and 1 (Pearson):

- $\lambda = 0 \rightarrow G^2 \approx 36.41$
- $\lambda = 0.5 \rightarrow \approx 32.17$
- $\lambda = 1 \rightarrow \approx 33.88$

All values suggest a strong deviation from independence and align with the Pearson result. All values lie between 32 and 36, indicating a clear departure from independence.

Interpretation of Question 1

- $\chi^2 \approx 33.88$ with $df=9$ yields $p \approx 0.0002 \rightarrow$ reject independence.
- Monte Carlo p-values consistently show significance as B increases.
- Cressie–Read divergence supports strong discrepancy from the null hypothesis.

R Code

Question 1: Chi-squared test and Cressie–Read

```
# Data
asbestos <- matrix(c(5,8,6,7,
                    9,13,12,7,
                    11,20,7,6,
                    4,6,6,4),
                  nrow=4, byrow=TRUE)
dimnames(asbestos) <- list(
  Y = c("Y1", "Y2", "Y3", "Y4"),
  G = c("G1", "G2", "G3", "G4")
)

# Standard chi-squared test
chisq.test(asbestos)

# Monte Carlo p-values for different B values
chisq.test(asbestos, simulate.p.value=TRUE, B=10)
chisq.test(asbestos, simulate.p.value=TRUE, B=100)
chisq.test(asbestos, simulate.p.value=TRUE, B=1000)
chisq.test(asbestos, simulate.p.value=TRUE, B=10000)

# Cressie–Read function
cressie_read <- function(observed, expected, lambda) {
  if (lambda == 0) {
    return(2 * sum(observed * log(observed / expected)))
  } else if (lambda == 1) {
```

```

    return(sum((observed - expected)^2 / expected))
  } else {
    ratio <- observed / expected
    term <- (observed / expected)^lambda - 1
    return((2 / (lambda * (lambda + 1))) * sum(expected * term))
  }
}

# Expected frequencies
expected <- chisq.test(asbestos)$expected
observed <- asbestos

# Compute statistics
cressie_read(observed, expected, 0) # λ = 0 (LLR)
cressie_read(observed, expected, 0.5) # λ = 0.5
cressie_read(observed, expected, 1) # λ = 1 (Pearson)

```

Question 2: Freeman–Tukey and Log-Likelihood Ratio with Bootstrapping:

- Compute two alternative statistics for the same 4×4 table:

1. Freeman–Tukey (FT):

$$FT = \sum_{i,j} (\sqrt{O_{ij}} - \sqrt{E_{ij}})^2.$$

2. Log-Likelihood Ratio (LLR or G^2):

$$G^2 = 2 \sum_{i,j} O_{ij} \ln(O_{ij}/E_{ij}).$$

Obtain p-values using approximation with $df = 9$.

Generate 1000 bootstrap tables using `r2dtable` to preserve margins.

- Recompute FT and LLR.
- Compare their p-values to the original table's p-value.

Determine the percentage of bootstrap tables with p-values the original p-value.

Results

Observed FT and LLR

FT_obs ≈ 11.275 , p-value ≈ 0.2598 (not significant)

LLR_obs ≈ 36.412 , p-value ≈ 0.00006 (highly significant)

Bootstrap with 1000 Replicates

Percentage of FT-bootstrap p-values $\leq$ original: 0%

Percentage of LLR-bootstrap p-values $\leq$ original: 0%

Bootstrap confirms that the original table is very extreme under H_0 .

Interpretation of Question 2

- Divergence measures can yield very **different conclusions** from the same data.
- **FT** is not significant, but **LLR** strongly rejects independence.
- Bootstrap confirms the **extremity** of the observed table.
- Supports results from Question 1 and shows LLR's **higher sensitivity** in certain cases.

R Code

Question 2: Freeman–Tukey, LLR, and Bootstrap

```
# --- Question 2: Freeman–Tukey, LLR, Bootstrap ---
```

```
# Use the asbestos 4×4 table
```

```
O <- matrix(c(5,8,6,7,
```

```
9,13,12,7,
```

```
11,20,7,6,
```

```
4,6,6,4),
```

```
nrow=4, byrow=TRUE)
```

```
r <- nrow(O)
```

```
c <- ncol(O)
```

```
row_sums <- rowSums(O)
```

```
col_sums <- colSums(O)
```

```
total <- sum(O)
```

```
E <- outer(row_sums, col_sums, "*") / total
```

```
# Freeman–Tukey (FT) statistic
```

```
freeman_tukey_stat <- function(O, E) {
```

```
  sum((sqrt(O) - sqrt(E))^2)
```

```
}
```

```
# Log-Likelihood Ratio (LLR =  $G^2$ )
```

```
llr_stat <- function(O, E) {
```

```
  mask <- (O > 0)
```

```
  2 * sum(O[mask] * log(O[mask] / E[mask])) }
```

```

# Function to get chi-squared p-value with df = (r-1)(c-1)
chisq_pvalue <- function(stat_val, r, c) {
  df <- (r - 1) * (c - 1)
  pchisq(stat_val, df = df, lower.tail = FALSE) }

# 1) Compute the observed FT and LLR and their p-values
FT_obs <- freeman_tukey_stat(O, E)
LLR_obs <- llr_stat(O, E)

pval_FT_obs <- chisq_pvalue(FT_obs, r, c)
pval_LLR_obs <- chisq_pvalue(LLR_obs, r, c)

cat("Observed Freeman-Tukey =", FT_obs, "; p-value =", pval_FT_obs, "\n")
cat("Observed LLR (G^2) =", LLR_obs, "; p-value =", pval_LLR_obs, "\n\n")

# 2) Generate 1000 bootstrap tables (keeping row & column margins)
set.seed(123)
B <- 1000
boot_tables <- r2dtable(n = B, r = row_sums, c = col_sums)

# 3) Compute p-values for each bootstrap table
FT_pvals <- numeric(B)
LLR_pvals <- numeric(B)

for(i in seq_len(B)) {
  O_star <- boot_tables[[i]]

```



```

E_star <- outer(rowSums(O_star), colSums(O_star), "*") / sum(O_star)

FT_star <- freeman_tukey_stat(O_star, E_star)
LLR_star <- llr_stat(O_star, E_star)

FT_pvals[i] <- chisq_pvalue(FT_star, r, c)
LLR_pvals[i] <- chisq_pvalue(LLR_star, r, c)
}

# 4) Compare to original table p-values
percent_FT <- mean(FT_pvals <= pval_FT_obs) * 100
percent_LLRL <- mean(LLR_pvals <= pval_LLRL_obs) * 100

cat("Percentage (FT bootstrap p-value <= original):", percent_FT, "%\n")
cat("Percentage (LLR bootstrap p-value <= original):", percent_LLRL, "%\n")

```

Question 3: Goodman–Kruskal’s Lambda

Tasks

- Suppose that there is an asymmetric relationship between the row variable (Years of Exposure) and the column variable (Asbestosis Grade).
- Compute Goodman–Kruskal’s $\lambda_{\text{row} \rightarrow \text{col}}$.
- Check the significance using a randomization approach with $B = 1000$.
- Interpret how grade helps in classification of exposure categories.

Definition of $\lambda_{\text{row} \rightarrow \text{col}}$

Goodman-Kruskal's Lambda is a Proportional Reduction in Error (PRE) measure:

$$\lambda_{\text{row} \rightarrow \text{col}} = \frac{\sum_j \max_i(O_{ij}) - \max_i(\sum_j O_{ij})}{\sum_{i,j} O_{ij} - \max_i(\sum_j O_{ij})}.$$

- $\max_i(\sum_j O_{ij})$ is the “best single row guess” if we do not know columns.
- $\sum_j \max_i(O_{ij})$ is the total “correct classification” if we pick the best row within each column.
- $\lambda_{\text{row} \rightarrow \text{col}}$ measures how much the column variable reduces misclassification in row classification.

$$\lambda_{\text{row} \rightarrow \text{col}} = (\sum_i \max_j(O_{ij}) - \max_j(\sum_i O_{ij})) / (\sum_{i,j} O_{ij} - \max_j(\sum_i O_{ij}))$$

It measures how much the column variable (grade) reduces misclassification in predicting the row variable (exposure).

Results

Observed λ :

From the observed data:

$$\lambda_{\text{row} \rightarrow \text{col}} \approx 0.43$$

This means there is approximately a 43% reduction in exposure classification error when grade is known.

Randomization Test ($B = 1000$)

We generated 1000 random tables preserving row and column sums using `r2dtable()`, and computed λ for each.

None of the bootstrap tables produced a λ as large as 0.43.

p-value $\approx 0 \rightarrow$ Strongly rejects the null hypothesis $H_0: \lambda = 0$.

Interpretation of Question 3

- $\lambda \approx 0.43$ shows a moderately strong predictive improvement.
- The randomization test confirms that this improvement is not due to chance.
- Therefore, there is a meaningful asymmetric relationship: grade helps predict exposure categories significantly better than random guessing.

R Code

Question 3: Goodman–Kruskal's Lambda (Manual)

```
# --- Question 3: Goodman–Kruskal's Lambda (Manual) ---
```

```
# 1) Define the 4x4 contingency table
```

```
asbestos <- matrix(c(5,8,6,7,
```

```
9,13,12,7,
```

```
11,20,7,6,
```

```
4,6,6,4),
```

```
nrow = 4, byrow = TRUE)
```

```
rownames(asbestos) <- c("Y1", "Y2", "Y3", "Y4")
```

```
colnames(asbestos) <- c("G1", "G2", "G3", "G4")
```

```
# A) Function to calculate GK Lambda (row/column)
```

```
gk_lambda_row_col <- function(M) {
```

```
  row_sums <- rowSums(M)
```

```

total <- sum(M)
best_guess <- max(row_sums)
correct_by_col <- sum(apply(M, 2, max))
(correct_by_col - best_guess) / (total - best_guess)
}

```

B) Compute observed lambda

```

lambda_obs <- gk_lambda_row_col(asbestos)
cat("Observed lambda =", lambda_obs, "\n")

```

C) Randomization/Bootstrap Test

```

gk_lambda_test <- function(M, B = 1000, seed = NULL) {
  if (!is.null(seed)) set.seed(seed)

```

```

  lam_obs <- gk_lambda_row_col(M)
  row_sums <- rowSums(M)
  col_sums <- colSums(M)
  lam_star <- numeric(B)

```

```

  for (i in seq_len(B)) {
    M_star <- r2dtable(1, row_sums, col_sums)[[1]]
    lam_star[i] <- gk_lambda_row_col(M_star)
  }

```

```

  p_val <- mean(lam_star >= lam_obs)
  return(list(lambda_observed = lam_obs,
    lambda_bootstrap = lam_star,
    p_value = p_val))

```

```
}  
  
# Example usage  
test_result <- gk_lambda_test(asbestos, B = 1000, seed = 123)  
cat("Observed lambda:", test_result$lambda_observed, "\n")  
cat("p-value from randomization:", test_result$p_value, "\n")
```

Overall Conclusions

- Question 1: The Pearson chi-squared test yields $\chi^2 \approx 33.88$, $df = 9$, with $p \approx 0.0002$. We reject the null hypothesis of independence with strong statistical confidence. Monte Carlo simulations show decreasing p-values with increasing B, confirming the result (down to $\sim 1 \times 10^{-4}$). Manual computation and Cressie–Read statistics ($\lambda = 0, 0.5, 1$) also support strong evidence of association between exposure years and asbestosis grade.
- Question 2: The Freeman–Tukey statistic is ≈ 11.28 with $p \approx 0.26$ (not significant). In contrast, the Log-Likelihood Ratio (LLR) is ≈ 36.41 with $p \approx 0.00006$, which is highly significant. A bootstrap test using 1000 samples found 0% of tables with p-values as small as the observed one, indicating the observed table is extreme under independence, especially from the LLR perspective.
- Question 3: Goodman–Kruskal’s $\lambda_{\text{row} \rightarrow \text{col}} \approx 0.43$. A randomization test with 1000 samples gave $p \approx 0$, meaning no permuted tables matched that level of predictive improvement. Knowing the asbestosis grade reduces misclassification of exposure bins by $\approx 43\%$, indicating a strong asymmetric and statistically significant relationship.

Thank you